Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (4)
BJT IV Characteristics

Spring 2023

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BJT IV Characteristics

Objectives

- To study the IV curve of a BJT.
- To experimentally evaluate the DC gain parameter of a BJT.
- To study the transfer function of a BJT circuit.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in active mode and find expressions for the collector current and the collector-emitter voltage.

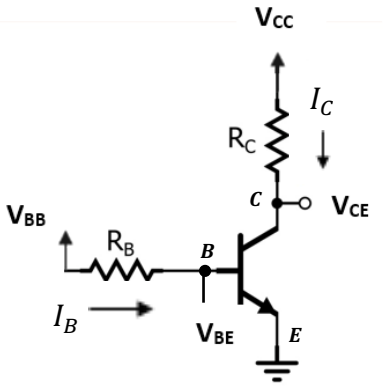


Figure 1: A BJT DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 8.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 5 kΩ (1), 10 kΩ (1), 100 kΩ (1)			
3	BJT NPN transistor (1)	2N3904	https://www.onsemi.com/pub/Collateral/2N3903-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): BJT IV Characteristics

1. Connect the circuit in figure 1 with R_C=10 kΩ and R_B=100 kΩ. Pay a special attention to how you connect the terminals of the BJT. Refer to the data sheet to identify the transistor terminals.

2. Set V_{BB} to a constant voltage of +10 V while connecting V_{CC} to a variable DC voltage source.
3. The IV characteristics (I_C versus V_{CE}) is obtained by slowly varying V_{CC} and measuring the voltage V_{CE} and V_{R_C} at each setting. Use table 2 to enter your results. Also record the measured values of V_{R_B} at each setting (to be used in part 2 of the experiment).

Part (2): BJT DC Gain Parameter (β)

1. The required data have been collected in part 1.

Part (3): BJT Transfer Function

1. Reconsider the circuit in figure 1.
2. Set V_{CC} to a constant voltage of +10 V while connecting V_{BB} to a variable DC voltage source.
3. The transfer function (V_{CE} versus V_{BB}) is obtained by slowly varying V_{BB} and measuring the voltage V_{CE} at each setting. Use table 3 to enter your results.
4. Repeat step 3 for another value of R_C, say 5 kΩ. Record your results in table 3.

Results and Analysis

Part (1): BJT IV Characteristics.

1. Calculate the currents I_C and I_B values in table 2 (Use $I_C = V_{RC}/R_C$ and $I_B = V_{RB}/R_B$).
2. Use graph 1 to sketch the IV curve.

Table 2. IV Characteristics Measurements.

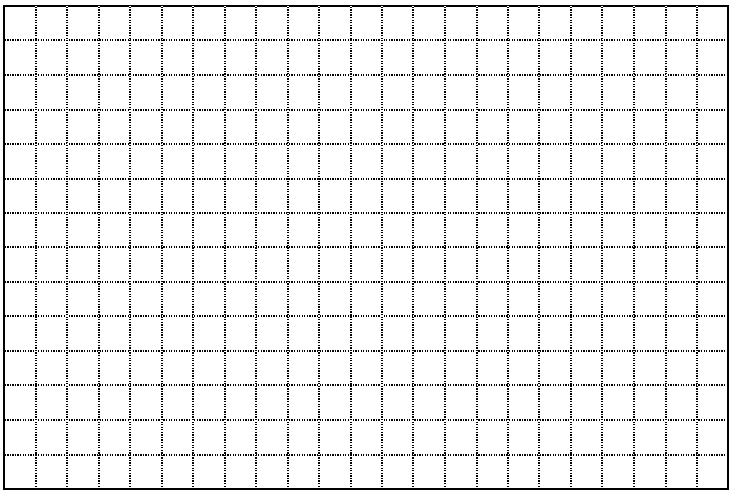
V_{CC} (V)	V_{CE} (V)	V_{R_C} (V)	I_C (mA)	V_{R_B} (V)	I_B (mA)	$\beta = I_C/I_B$
0.0						
1.0						
2.0						
3.0						
4.0						
5.0						
6.0						
7.0						
8.0						
9.0						
10.0						

3. Comment on the IV curve that you obtained. Can you identify the transistor modes of operation? Show your answer on the curve.

Answer:

4. Is the collector current constant in the active mode region of the IV curve? Find an approximate value of the Early voltage.

Answer:



Graph 1: Transistor IV Curve.

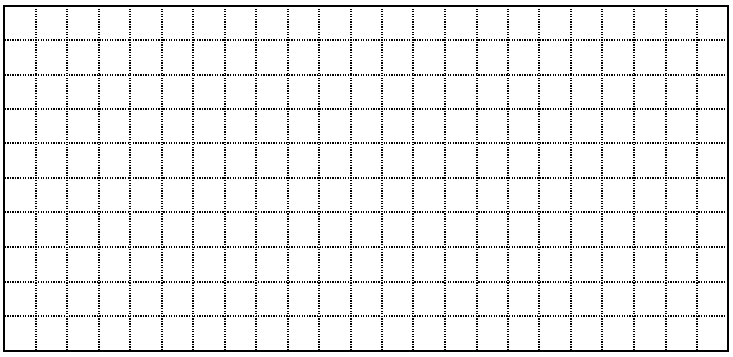
Part (2): BJT DC Gain Parameter (β)

1. Calculate the current I_B values in table 2. Also, calculate the DC gain parameter (β).
2. Why is the value of β not constant in the table? Explain in details.

Answer:

3. Use graph 2 to sketch β vs. I_C relation. Compare to the curve given in the transistor data sheet and give your comments.

Answer:



Graph 2: DC gain parameter.

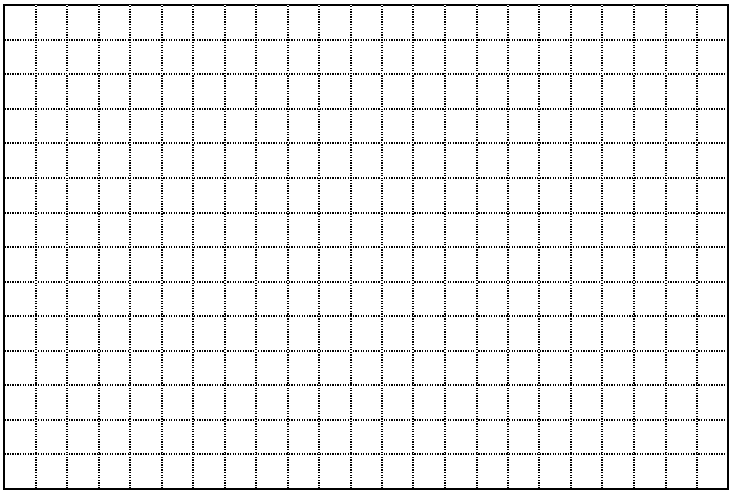
Part (3): BJT Transfer Function

1. Use graph 3 to sketch the two transfer functions using the data in table 3, for both R_C values.
2. Can you identify the transistor modes of operation? Show your answer on the curves.

Answer:

Table 3. Transfer Function Measurements.

	R_C = 10 kΩ	R_C = 5 kΩ
V_{BB} (V)	V_{CE} (V)	V_{CE} (V)
0.0		
1.0		
2.0		
3.0		
4.0		
5.0		
6.0		
7.0		
8.0		
9.0		
10.0		



Graph 3: Circuit transfer function.

3. Comment on the general shape of the transfer function. In particular, explain the main differences between the two curves, in relation to R_C value.

Answer:

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

2N3904
Silicon NPN Transistor
General Purpose
TO–92 Type Package

Absolute Maximum Ratings:

Collector–Emitter Voltage, V_{CEO} 40V
Collector–Base Voltage, V_{CB} 60V
Emitter–Base Voltage, V_{EBO} 6V
Continuous Collector Current, I_C 200mA
Total Device Dissipation (T_A = +25°C), P_D 625mW
Derate Above 25°C . . . . . 2.8mW/°C
Total Device Dissipation (T_C = +25°C), P_D 1.5W
Derate Above 25°C . . . . . 12mW/°C
Operating Junction Temperature Range, T_J . . . . . –55° to +150°C
Storage Temperature Range, T_{stg} . . . . . –55° to +150°C
Thermal Resistance, Junction to Case, R_{thJC} . . . . . 83.3°C/W
Thermal Resistance, Junction to Ambient, R_{thJA} . . . . . 200°C/W

Electrical Characteristics: (T_A = +25°C unless otherwise specified)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
OFF Characteristics						
Collector–Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0, Note 1	40	–	–	V
Collector–Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10μA, I_E = 0	60	–	–	V
Emitter–Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10μA, I_C = 0	6	–	–	V
Collector Cutoff Current	I_{CEX}	V_{CE} = 30V, V_{EB} = 3V	–	–	50	nA
Base Cutoff Current	I_{BL}	V_{CE} = 30V, V_{EB} = 3V	–	–	50	nA
ON Characteristics (Note 1)						
DC Current Gain	h_{FE}	V_{CE} = 1V, I_C = 0.1mA	40	–	–	
		V_{CE} = 1V, I_C = 1mA	70	–	–	
		V_{CE} = 1V, I_C = 10mA	100	–	300	
		V_{CE} = 1V, I_C = 50mA	60	–	–	
		V_{CE} = 1V, I_C = 100mA	30	–	–	

Note 1. Pulse Test: Pulse Width ≤ 300μs, Duty Cycle ≤ 2%.

Electrical Characteristics (Cont'd): (T_A = +25°C unless otherwise specified)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
ON Characteristics (Cont'd) (Note 1)						
Collector–Emitter Saturation Voltage	$V_{CE(sat)}$	I_C = 10mA, I_B = 1mA	–	–	0.2	V
		I_C = 50mA, I_B = 5mA	–	–	0.3	V
Base–Emitter Saturation Voltage	$V_{BE(sat)}$	I_C = 10mA, I_B = 1mA	0.65	–	0.85	V
		I_C = 50mA, I_B = 5mA	–	–	0.95	V
Small–Signal Characteristics						
Current Gain–Bandwidth Product	f_T	I_C = 10mA, V_{CE} = 20V, f = 100MHz	300	–	–	MHz
Output Capacitance	C_{obo}	V_{CB} = 5V, I_E = 0, f = 1MHz	–	–	4.0	pF
Input Capacitance	C_{ibo}	V_{CB} = 0.5V, I_C = 0, f = 1MHz	–	–	8.0	pF
Input Impedance	h_{ie}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	1.0	–	10	kΩ
Voltage Feedback Ratio	h_{re}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	0.5	–	8.0	x 10^{-4}
Small–Signal Current Gain	h_{fe}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	100	–	400	
Output Admittance	h_{oe}	I_C = 1mA, V_{CE} = 10V, f = 1kHz	1.0	–	30	μmhos
Noise Figure	NF	I_C = 100μA, V_{CE} = 5V, R_S = 1kΩ, f = 10Hz to 15.7kHz	–	–	5.0	db
Switching Characteristics						
Delay Time	t_d	V_{CC} = 3V, V_{EB} = 0.5V, I_C = 10mA, I_{B1} = 1mA	–	–	35	ns
Rise Time	t_r		–	–	35	ns
Storage Time	t_s	V_{CC} = 3V, I_C = 10mA, I_{B1} = I_{B2} = 1mA	–	–	200	ns
Fall Time	t_f		–	–	50	ns

Note 1. Pulse Test: Pulse Width ≤ 300μs, Duty Cycle ≤ 2%.